
OFBD: Object-Focused Background Debiasing for Long-Tailed Learning

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Abstract

Balancing performance trade-offs on long-tailed data distributions remains a long-standing challenge in visual recognition. Existing methods mainly improve tail classes through re-balancing, representation learning, or data augmentation, but the underlying cause of tail class degradation is still insufficiently explored. In this paper, we find that standard long-tailed training induces background-biased representation and optimization: tail classes suffer larger background distribution shifts and become increasingly driven by background gradients. This reveals that tail degradation is not merely caused by insufficient samples, but also by the learning of irrelevant background features. To tackle this issue, we propose Object-Focused Background Debiasing (OFBD), a framework that mitigates background bias from both distribution and optimization perspectives. Specifically, Foreground-guided CutMix preserves target-related foregrounds while diversifying complementary backgrounds, and Background-guided Feature Rectification suppresses background-biased features without learnable parameters or additional training. Extensive experiments show that our method improves overall accuracy, achieves significant tail-class gains, and can serve as a plug-in for mainstream long-tailed methods without external data or pretrained recognition models. The code is available at: <https://ofbd-neurips2026-longtail-learning.github.io/>

1 Introduction

The prevailing paradigms in visual recognition [37, 80, 16] owe much of their success to large-scale, carefully curated datasets [66, 47] with balanced class distributions. However, real-world data naturally exhibits highly imbalanced or long-tailed distributions [97]. When optimized on such skewed distributions, standard training protocols inevitably become biased toward data-abundant head classes, resulting in severe performance degradation on data-scarce tail classes.

Existing methods addressing the class imbalance issue mainly fall into three paradigms. First, class re-balancing strategies such as resampling [50, 60] or reweighting [21, 11], amplify tail signals but may overfit tail classes [74]. Second, architectural improvements correct feature and classifier bias via contrastive representation learning [110, 83, 32]; and via logit calibration [53] or expert routing [87, 9, 100]. Third, information augmentation techniques synthesize or transfer semantic variations for tail classes [84, 70], beyond conventional augmentations [1] (*e.g.*, cropping, flipping). Despite these improvements, existing methods mainly compensate for data quantity, leaving the degradation of tail representations during standard optimization [40] largely unexplored.

*This work was done while Shenghan Chen was a visiting student at Westlake University.

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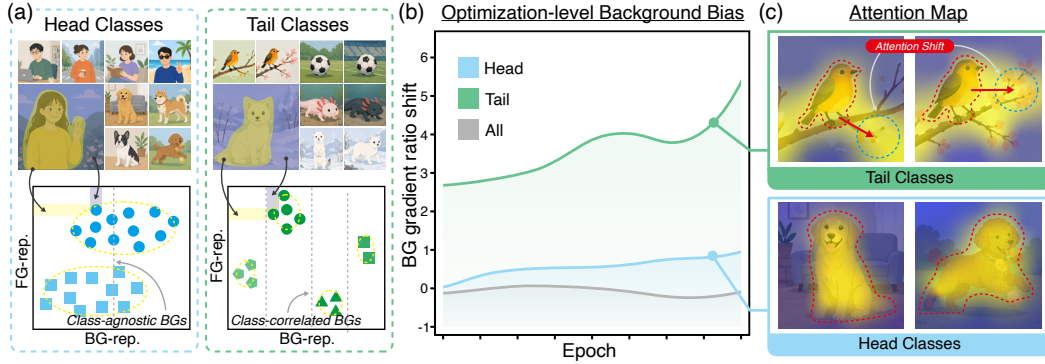


Figure 1: Motivation of OFBD. Tail-class degradation stems from both sample scarcity and background bias in distribution and optimization. (a) Head classes possess diverse, class-agnostic backgrounds (BGs), making background representations (BG-Rep.) uninformative for classification. Conversely, tail classes often co-occur with class-correlated BGs (e.g., snow fox in snowy scenes), which makes BG-Rep. a predictive shortcut and encourages reliance on BG cues over sparse foregrounds. Gray dashed lines denote BG-based decision boundaries. (b) Optimization-level background bias. “Head” and “Tail” represent the evaluation results under the long-tailed setting, while “All” indicates the overall result on the full balanced dataset. (c) Attention maps reveal that tail predictions co-activate foregrounds and backgrounds, whereas head predictions remain object-focused.

Interestingly, recent advances in self-supervised [28, 49] and few-shot learning [88] demonstrate that models can learn highly transferable representations from extremely limited data [15]. Such findings indicate that sample scarcity alone is insufficient to explain the profound representational collapse observed in tail classes. This paradox raises a critical question: **If limited data is not an inherent barrier, why do long-tailed models fail to capture the discriminative features of tail classes?**

To investigate the underlying mechanism, we visualize the spatial representations learned by representative long-tailed networks [53, 110]. As illustrated in Fig. 1(c), we observe a striking phenomenon: standard imbalanced training severely compromises the spatial inductive bias of the network, causing it to co-activate target foregrounds with irrelevant background regions. This empirical evidence reveals that tail-class degradation extends beyond simple data scarcity; crucially, it induces a profound background bias in both representation and optimization. Consequently, discriminative foreground features are marginalized by spurious contextual cues, leading to degradation on tail classes [77, 55]. Therefore, resolving the long-tailed dilemma necessitates a dual approach: enhancing target-related foreground while explicitly suppressing the exploitation of irrelevant background evidence.

Suppressing irrelevant background evidence requires understanding its entanglement with targets. Building upon recent insights [55, 73], we attribute this background bias to two structural forms of input-level co-occurrence: object-scene (e.g., fish with water) and object-object (e.g., racket with person) entanglements (Fig. 1(a)). These co-occurrences bias optimization toward spurious background shortcuts, resulting in background-biased representations and skewed BG feature distributions (Fig. 1(b)). We formalize this contextual and structural analysis in Sec. 3. Although spatial mixup strategies like CutMix [98] can disrupt these co-occurrences, prior works have shown that naive random replacements may introduce irrelevant noise [58, 29]. To tackle this issue, we propose Object-Focused Background Debiasing (OFBD), a framework that mitigates background bias from both distribution and optimization perspectives. Specifically, *Foreground-guided CutMix* (FG-CutMix) uses a reinforcement-learning selector to handle discrete and non-differentiable region selection, preserving target-related foregrounds while replacing backgrounds to break foreground-background co-occurrence. Meanwhile, *Background-guided Feature Rectification* (BFR) estimates background contribution from channel-wise feature statistics and down-weights background-biased spatial features in a parameter-free manner, avoiding head-class-dominated learnable rectification.

Overall, the main contributions of this paper are summarized as follows:

- **Novel Perspective:** We systematically investigate the inherent background bias in long-tailed recognition. Through structural analysis and empirical validation, we reveal how this bias shifts feature distributions and misguides optimization dynamics.

- **Dual Debiasing Framework:** We propose FG-CutMix and BFR to address the two biases. FG-CutMix integrates reinforcement learning [71] to preserve target foregrounds, while BFR rectifies features to reduce background reliance in a parameter-free manner.
- **Superior Performance & Plug-and-Play Versatility:** Extensive experiments demonstrate that without relying on external data or pretrained vision models, OFBD significantly boosts tail-class performance and overall accuracy. Furthermore, it inherently serves as a highly efficient plug-in module for existing mainstream long-tailed models.

2 Related Work

Long-tailed Learning: Real-world data are often long-tailed [11, 111, 57, 45, 21]. Existing methods include class re-balancing [21, 102, 50, 65, 75], information augmentation [70, 41, 48, 91, 104], and architectural improvements [24, 13, 68, 93, 76]. Yet they mainly compensate class quantity: re-balancing changes sample/loss weights, augmentation adds diversity without separating foreground features from background features, and architectures rarely rectify feature-level background bias. We instead mitigate distribution- and optimization-level background bias by decoupling target-related foreground features from spurious background features.

Background Bias in Visual Recognition: Deep neural networks are notoriously prone to background bias, often relying on spurious contextual correlations (shortcuts) rather than intrinsic object semantics [18, 17, 95, 56]. While various debiasing strategies have been proposed [54, 82, 34, 38, 5], they predominantly assume balanced data distributions, leaving their behavior under severe class imbalance unexplored. Our work bridges this critical gap by revealing that standard long-tailed optimization inherently exacerbates these background biases. In response, we introduce a dual debiasing framework, offering a principled approach to robust and unbiased long-tailed recognition.

RL-driven Data Augmentation: Reinforcement Learning (RL) has been used to search augmentation policies [10, 46, 84], select transformation operations or magnitudes [19, 12], and identify informative regions [89, 90, 44]. These methods optimize validation performance, augmentation strength, or region informativeness for general robustness, but are not tailored to long-tailed recognition, where tail classes are more vulnerable to spurious background reliance. In contrast, our RL selector is designed for FG-CutMix under long-tailed training. By selecting foreground regions that preserve target semantics after background replacement, the selector disrupts foreground-background co-occurrence and reduces tail-class background reliance.

3 Analysis and Motivation

In this section, we analyze background bias from two complementary perspectives and provide empirical evidence. At the distribution level, we study how the learned background feature distribution deviates from that of the full balanced dataset. At the optimization level, we examine how training becomes increasingly driven by background-related gradients.

3.1 Foreground-Background Decomposition

Following prior works [72, 61, 86], we decompose an input image x into the target-related foreground and the target-irrelevant background, including co-occurring scenes and objects. Let $F = \Phi(x) \in \mathbb{R}^{D \times H \times W}$ be the feature map extracted by the encoder. Since foreground and background are spatial concepts, we first separate foreground and background regions on F according to the decomposition of x , and then aggregate them into f_{fg} and f_{bg} , respectively. The image-level features can be written as $f = f_{fg} + f_{bg}$, where f_{fg} and f_{bg} denote foreground and background features.

3.2 Distribution-level Background Bias

For each class y , we use p_y and q_y to denote the background distributions under long-tailed and full-dataset training, respectively. The corresponding background feature means are $\mu_{bg}(y) = \mathbb{E}_{b \sim p_y}[f_{bg}]$ and $\bar{\mu}_{bg}(y) = \mathbb{E}_{b \sim q_y}[f_{bg}]$. We then define the background bias of class y as $\Delta D_{bg}(y) := \mu_{bg}(y) - \bar{\mu}_{bg}(y)$, which measures the shift of the learned background feature mean from the full-dataset background feature mean. Since tail classes have far fewer samples than head classes, their

empirical background distributions are more likely to deviate from q_y . Accordingly, we assume $\|p_y^{\text{tail}} - q_y\|_1 > \|p_y^{\text{head}} - q_y\|_1$, which yields a larger background distribution shift for tail classes:

$$\|\Delta D_{bg}^{\text{tail}}\|_1 > \|\Delta D_{bg}^{\text{head}}\|_1. \quad (1)$$

3.3 Optimization-level Background Bias

We next consider background bias from the optimization perspective. Let g_t denote the gradient signal at epoch t . Consistent with the feature decomposition above, we first separate foreground and background regions on F , and then aggregate their gradient magnitudes into $g_{fg,t}$ and $g_{bg,t}$, respectively. We define the background gradient ratio and its shift for class y as:

$$R_t(y) = \frac{G_{bg,t}(y)}{G_{fg,t}(y) + G_{bg,t}(y)}, \quad R_0(y) = \frac{1}{T_0} \sum_{t=1}^{T_0} R_t(y), \quad \Delta R_t(y) = R_t(y) - R_0(y), \quad (2)$$

where $G_{fg,t}(y) = \|g_{fg,t}(y)\|_1$ and $G_{bg,t}(y) = \|g_{bg,t}(y)\|_1$. Since the absolute value of $R_t(y)$ can be affected by the initial foreground-background composition of each class, $\Delta R_t(y)$ better captures whether optimization progressively leans toward background features. Under long-tailed training, tail classes receive weaker foreground supervision, while background cues are easier to exploit and continue to contribute to the loss. We therefore assume:

$$\Delta R_t^{\text{tail}} > \Delta R_t^{\text{head}}. \quad (3)$$

Detailed derivations and assumptions motivating these hypotheses (Eq. 1 and Eq. 3) are provided in the Appendix B. We empirically validate these claims below.

3.4 Empirical Evidence

To validate the above analysis, we empirically examine background bias from both distribution and optimization perspectives. Visualized results and experimental details are provided in Appendix D.3.

Observation 1: Tail classes suffer larger distribution-level shift. Fig. 5a shows larger tail-class background distribution shift than head classes due to limited samples and less diverse backgrounds, motivating FG-CutMix to diversify backgrounds while preserving target foregrounds.

Observation 2: Long-tailed training amplifies optimization-level bias. Fig. 5b shows a larger background-gradient-ratio increase under long-tailed training, especially for tail classes. This indicates that insufficient and imbalanced foreground supervision shifts optimization toward easily exploitable background features, motivating BFR to suppress background-biased gradients.

Observation 3: Tail-class background influence is more unstable. Fig. 5b further shows stronger fluctuations in tail-class background-gradient ratios, motivating parameter-free BFR and the reference selector in FG-CutMix for stable debiasing and foreground selection.

4 Method

Guided by Sec. 3, our framework addresses background bias from two aspects. FG-CutMix targets the distribution-level shift ΔD_{bg} by preserving foregrounds while replacing backgrounds. BFR targets the optimization-level shift ΔR_t by down-weighting background-biased features. Together, they mitigate biased background distributions and background-driven optimization in long-tailed training. We provide a theoretical interpretation of the effectiveness of both modules in Appendix B.4.

4.1 Foreground-guided CutMix

Unlike naive CutMix, which may exacerbate bias by sampling background-only regions [55], our Foreground-guided CutMix (FG-CutMix) explicitly preserves target foregrounds. Furthermore, its RL-based selector introduces minimal computational overhead, ensuring practicality for large-scale training. Detailed experiments evaluating the computational cost are provided in Appendix D.8.

4.1.1 RL-based Foreground Region Selection

The key challenge in FG-CutMix is selecting foreground regions effective for background replacement. CAM-based [106, 31] and SAM-based [35, 64] strategies can localize complete foregrounds, but accurate localization may not yield the best mixing region. Since CutMix benefits from partial-view recognition and regional perturbation [98], effective regions should balance target preservation and background replacement, as overly complete regions may weaken background variation and regularization. Since foreground-region selection is discrete and non-differentiable, we use an RL-based selector to learn an adaptive policy, capture class-specific foreground importance.

RL-based foreground candidate generation. Given the feature map $F = \Phi(x) \in \mathbb{R}^{D \times H \times W}$, we first randomly generate a region proposal set $\mathcal{P}(x) = \{b_1, \dots, b_N\}$ with different locations and scales. Each proposal b_i corresponds to a candidate crop region on the input image. Instead of ranking these proposals by local energy [85, 26], we introduce an RL-based region selector π_θ to select K proposals from $\mathcal{P}(x)$ as foreground candidates. Specifically, π_θ predicts a selection probability over $\mathcal{P}(x)$, and K proposals are sampled without replacement to form the foreground candidate pool (FG candidate pool) $\mathcal{R}_\theta(x) = \{r_1, \dots, r_k, \dots, r_K\}$. This allows the selector to learn which randomly generated regions are more suitable for preserving target-related foregrounds in FG-CutMix.

Reward design. Given the selected candidate pool, we construct mixed samples $\{\tilde{x}_1, \dots, \tilde{x}_K\}$ and obtain class prediction probabilities $p_k \in [0, 1]^C$ from the classifier. Let y and y^{bg} denote the target label and the background-source label, respectively. The reward for candidate r_k is defined as:

$$\rho_k = \mathbf{1}(p_k[y] > p_k[y^{\text{bg}}]). \quad (4)$$

Here, p_k provides the reward signal and $o_k = \pi_\theta(r_k | F)$ denotes the selector confidence, encouraging regions that preserve target semantics over background-induced semantics.

Policy learning. The region selector π_θ defines a selection distribution over the proposal set $\mathcal{P}(x)$ and samples K proposals $\mathcal{R}_\theta(x)$ as foreground candidates. Since proposal selection is discrete and non-differentiable, we optimize π_θ via policy gradient [36, 69, 94]. Since selected candidates within each image are not directly comparable due to varying classifier confidence [101] and potential background bias, we follow GRPO [71] and normalize rewards within each candidate pool. Since tail-class background gradients fluctuate more (Sec. 3.4), we use the previous-epoch selector as a frozen reference π_{ref} to stabilize foreground selection. The policy learning objective is:

$$\begin{aligned} \mathcal{L}_{\text{RL}} &= -\frac{1}{K} \sum_{k=1}^K [\min(\hat{s}_{1,k} A_k, \hat{s}_{2,k} A_k) - \beta \mathcal{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) - \alpha \text{CE}(o_k, \rho_k)], \\ \mathcal{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) &= \frac{\pi_{\text{ref}}(r_k | F)}{\pi_\theta(r_k | F)} - \log \frac{\pi_{\text{ref}}(r_k | F)}{\pi_\theta(r_k | F)} - 1, \\ \hat{s}_{1,k} &= \frac{\pi_\theta(r_k | F)}{\pi_{\text{ref}}(r_k | F)}, \quad \hat{s}_{2,k} = \text{clip} \left(\frac{\pi_\theta(r_k | F)}{\pi_{\text{ref}}(r_k | F)}, 1 - \epsilon_c, 1 + \epsilon_c \right), \\ A_k &= \frac{\rho_k - \text{mean}(\rho_1, \dots, \rho_K)}{\text{std}(\rho_1, \dots, \rho_K) + \epsilon}, \quad o_k = \pi_\theta(r_k | F), \end{aligned} \quad (5)$$

where θ denotes the parameters of the region selector π_θ , ϵ_c clamps the policy ratio, and β controls the KL penalty. Following prior works [69, 71], the frozen reference selector π_{ref} and KL term stabilize policy learning by limiting abrupt policy shifts, while the clipped objective prioritizes high-advantage foreground candidates without overly large updates. The auxiliary term, weighted by α , encourages region selector confidence to align with the reward signal.

4.1.2 FG-CutMix Sample Construction

Given a selected foreground candidate $r_k \in \mathcal{R}(x)$, we preserve the target-related region in x and replace its complement with content from another image x' . Let M_k be the binary mask of r_k , where $M_k = 1$ denotes the preserved region, and $M_k = 0$ denotes the complement. The mixed sample is $\tilde{x}_k := M_k \odot x + (1 - M_k) \odot x'$, where $\odot$ denotes element-wise multiplication. This preserves target foregrounds, varies surrounding backgrounds, and reduces foreground-background co-occurrence. Following prior works [58, 51], we correct the mixed label as $\tilde{y}_k$ for reliable supervision.

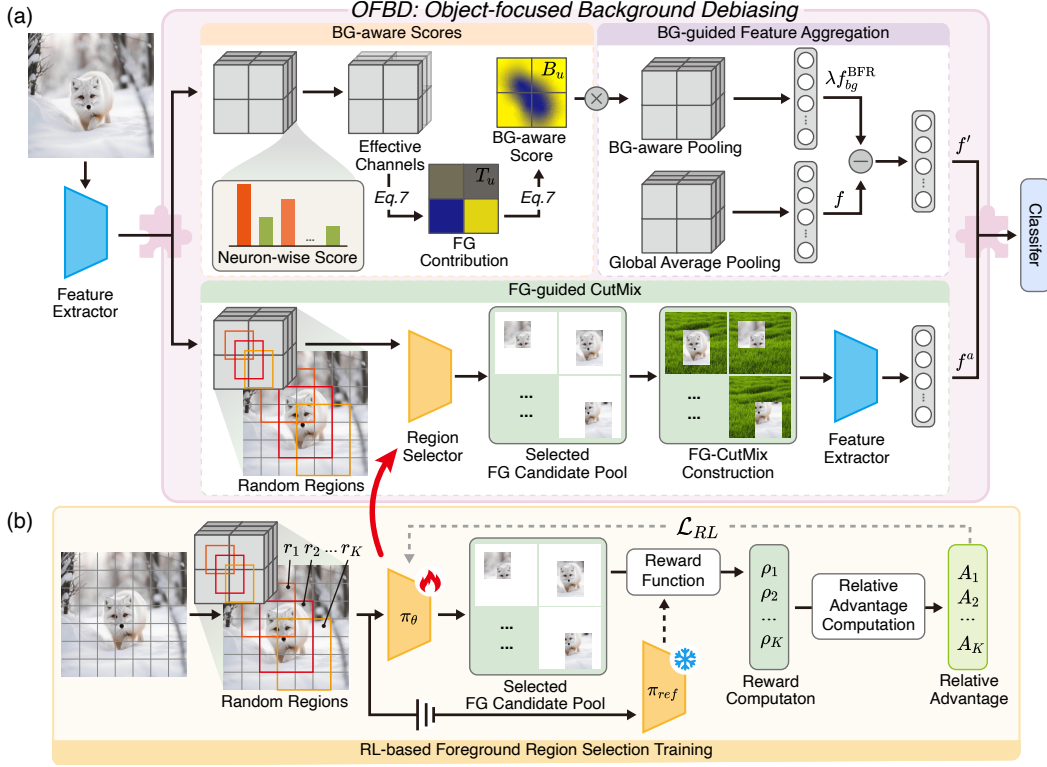


Figure 2: Overview of OFBD. The pink box denotes OFBD as a plug-in module for mainstream long-tailed models. (a) FG-CutMix selects target-related foregrounds from random regions and replaces complementary backgrounds, while BFR estimates background-aware scores to rectify features before classification. (b) The region selector π_θ is optimized via reinforcement learning, and a frozen reference selector π_{ref} stabilizes the training of the selector.

4.2 Background-guided Feature Rectification

Although FG-CutMix alleviates distribution-level background bias by restoring background diversity, optimization may still favor irrelevant background cues that are easier to exploit than sparse tail-class foreground evidence. We therefore introduce Background-guided Feature Rectification (BFR), which uses background-aware scores to down-weight biased feature locations in the final representation. To avoid head-class-dominated gradients from learnable parameters, BFR is strictly parameter-free. Inspired by SimAM [96] and LaSt-ViT [73], BFR estimates foreground contribution from channel-wise feature statistics: larger normalized deviations from stable channel means indicate higher foreground contribution. Effective channel aggregation further prevents sparse tail-class foreground channels from being diluted and derives background-aware scores for rectification.

4.2.1 Background-aware Scores

Given the feature map $F = \Phi(x) \in \mathbb{R}^{D \times H \times W}$, let u denote a spatial location and $F_{d,u}$ denote the activation at channel d and location u , we compute a neuron-wise score $S_{d,u}$:

$$S_{d,u} = \text{sigmoid} \left(\frac{(F_{d,u} - \mu_d)^2}{4(\sigma_d^2 + \epsilon)} \right), \quad \bar{S}_u = \frac{1}{D} \sum_{d=1}^D S_{d,u}, \quad (6)$$

where μ_d and σ_d^2 are the mean and variance of channel d over spatial locations, respectively. The detailed derivation is provided in Appendix B.3. We then retain effective channels $\mathcal{C}_u = \{d \mid S_{d,u} \geq \bar{S}_u\}$ and compute the foreground contribution T_u and background-aware score B_u as:

$$T_u = \frac{1}{|\mathcal{C}_u|} \sum_{d \in \mathcal{C}_u} S_{d,u}, \quad B_u = \text{sigmoid} \left(\frac{\bar{T} - T_u}{\sigma_T + \epsilon} \right), \quad (7)$$

where $\bar{T}$ and σ_T are the mean and standard deviation of T_u over all feature locations, respectively. A larger B_u indicates stronger background bias at location u .

Table 1: Comparisons on CIFAR-LT [4] datasets with varying imbalance ratios (r). Results are reported across all classes as well as different groups based on training sample sizes ("Many", "Med", and "Few"). For each method, the first row reports the original baseline accuracy, and the second row shows the performance after applying OFBD.

Method	CIFAR-100-LT			CIFAR-10-LT			Statistic($r = 100$)		
	$r=100\uparrow$	$r=50\uparrow$	$r=10\uparrow$	$r=100\uparrow$	$r=50\uparrow$	$r=10\uparrow$	Many $\uparrow$	Med. $\uparrow$	Few $\uparrow$
CE	38.32	43.94	55.71	70.45	74.82	86.44	65.19	37.06	9.14
+OFBD	44.83 $\pm$ 6.51	47.79 $\pm$ 3.85	58.22 $\pm$ 2.51	74.33 $\pm$ 3.88	78.68 $\pm$ 3.86	87.30 $\pm$ 0.86	65.48 $\pm$ 0.29	42.09 $\pm$ 5.03	23.93 $\pm$ 14.79
LDAM-DRW [4] (NeurIPS'19)	42.09	46.57	58.64	77.06	81.09	88.02	61.51	41.67	20.22
+OFBD	45.81 $\pm$ 3.72	49.55 $\pm$ 2.98	59.09 $\pm$ 0.45	79.61 $\pm$ 2.55	82.43 $\pm$ 1.34	88.76 $\pm$ 0.74	64.93 $\pm$ 3.42	45.81 $\pm$ 4.14	26.61 $\pm$ 6.39
BBN [107] (CVPR'20)	42.45	47.13	59.23	79.78	81.24	88.32	42.63	50.70	32.60
+OFBD	44.81 $\pm$ 2.36	48.32 $\pm$ 1.19	60.57 $\pm$ 1.34	80.69 $\pm$ 0.91	84.41 $\pm$ 3.17	89.28 $\pm$ 0.96	43.94 $\pm$ 1.31	51.83 $\pm$ 1.13	37.64 $\pm$ 5.04
BCL [110] (CVPR'22)	51.84	56.57	64.03	84.31	87.21	90.92	66.85	52.54	33.07
+OFBD	54.19 $\pm$ 2.35	57.94 $\pm$ 1.37	64.91 $\pm$ 0.88	86.34 $\pm$ 2.03	88.12 $\pm$ 0.91	91.63 $\pm$ 0.71	66.79 $\pm$ 0.06	53.46 $\pm$ 0.92	40.34 $\pm$ 7.27
SBCL [22] (CVPR'23)	44.93	48.65	57.83	74.84	80.05	84.48	64.35	45.27	22.14
+OFBD	46.92 $\pm$ 1.99	49.08 $\pm$ 0.43	58.44 $\pm$ 0.61	76.52 $\pm$ 1.68	81.43 $\pm$ 1.38	84.89 $\pm$ 0.41	65.26 $\pm$ 0.91	45.35 $\pm$ 0.08	27.36 $\pm$ 5.22
GBG [43] (AAAI'24)	51.92	56.92	64.32	85.12	87.64	91.23	66.94	53.15	32.96
+OFBD	54.23 $\pm$ 2.31	58.37 $\pm$ 1.45	64.98 $\pm$ 0.66	86.57 $\pm$ 1.45	88.25 $\pm$ 0.61	91.61 $\pm$ 0.38	66.81 $\pm$ 0.13	53.57 $\pm$ 0.42	40.32 $\pm$ 7.36
MKP [8] (CVPR'26)	53.21	57.63	68.74	86.31	88.26	92.53	67.28	54.86	34.91
+OFBD	54.42 $\pm$ 1.21	58.41 $\pm$ 0.78	68.12 $\pm$ 0.62	86.78 $\pm$ 0.47	88.42 $\pm$ 0.16	92.02 $\pm$ 0.51	67.57 $\pm$ 0.29	55.95 $\pm$ 1.09	37.29 $\pm$ 2.38

4.2.2 Background-guided Feature Aggregation

Based on the background-aware score B_u , we estimate and remove the background-related component from the image-level feature. Let f denote the standard image-level feature:

$$f = \frac{1}{HW} \sum_u F_u, \quad f_{bg}^{BFR} = \frac{1}{HW} \sum_u B_u F_u. \quad (8)$$

Here, f_{bg}^{BFR} denotes the estimated background-biased component. The rectified feature is then obtained by removing this estimated background-biased feature:

$$f' = f - \lambda f_{bg}^{BFR} = \frac{1}{HW} \sum_u (1 - \lambda B_u) F_u, \quad (9)$$

where λ controls the strength of background rectification. In this way, BFR softly down-weights background-biased feature locations instead of relying on a hard foreground-background mask.

Notably, BFR introduces no learnable parameters and **does not require additional training**, making it less affected by label imbalance in long-tailed recognition. By relying on feature statistics rather than extra supervision, BFR provides a lightweight way to rectify background-biased features.

4.3 Training Objective

For a given training sample (x, y) , FG-CutMix generates K augmented samples and aggregates them into a single feature f^a with target label $\tilde{y}$. Meanwhile, BFR produces the rectified feature f' for the original sample x . The overall objective is:

$$\mathcal{L} = \mathcal{L}_{cls} + \gamma \mathcal{L}_{RL}, \quad \mathcal{L}_{cls} = \frac{1}{2} [\ell_{LT}(h(f^a), \tilde{y}) + \ell_{LT}(h(f'), y)], \quad (10)$$

where $h(\cdot)$ is the classifier head, ℓ_{LT} denotes the underlying long-tailed objective, $\mathcal{L}_{RL}$ is the reinforcement learning loss (Eq. 5), and γ controls the strength of policy learning.

5 Experiments

5.1 Experiment Setup

Datasets and Metrics. Our proposed framework is evaluated on four long-tailed benchmarks: CIFAR-10-LT [4], CIFAR-100-LT [4], ImageNet-LT [52], and iNaturalist 2018 [79]. Following standard evaluation protocols [84, 105], we report Top-1 accuracy. For CIFAR datasets, we evaluate effectiveness of OFBD across three imbalance ratios ($r \in \{100, 50, 10\}$) and decompose results into "Many" (> 100 samples), "Med." ($20 \sim 100$), and "Few" (< 20) categories for further analysis. For ImageNet-LT and iNaturalist 2018, we assess various backbones to verify architecture generality.

Compared Methods. To demonstrate the effectiveness and plug-and-play versatility of OFBD, we integrate it into representative, open-source long-tailed methods to ensure reproducibility. These span

Table 2: Top-1 accuracy on ImageNet-LT [52] and iNaturalist 2018 [79].

Method	ImageNet-LT	iNature2018
<i>ResNet-50 [20] backbone</i>		
CE	44.80	65.95
+OFBD	47.39 $\uparrow$ 2.59	69.76 $\uparrow$ 3.81
BCL [110] (CVPR'22)	55.67	71.81
+OFBD	56.81 $\uparrow$ 1.14	72.34 $\uparrow$ 0.53
GCL [39] (CVPR'22)	55.46	70.76
+OFBD	56.34 $\uparrow$ 0.88	71.92 $\uparrow$ 1.16
GBG [43] (AAAI'24)	57.13	71.92
+OFBD	57.64 $\uparrow$ 0.51	72.36 $\uparrow$ 0.44
MKP [8] (CVPR'26)	57.92	74.41
+OFBD	58.51 $\uparrow$ 0.59	74.63 $\uparrow$ 0.22
<i>ViT-B [14] backbone</i>		
ViT [14] (ICLR'21)	37.51	54.23
+OFBD	39.17 $\uparrow$ 1.66	56.62 $\uparrow$ 2.39
DeiT-LT [63] (CVPR'24)	55.64	72.92
+OFBD	56.21 $\uparrow$ 0.57	74.56 $\uparrow$ 1.64

Table 3: Ablation study on main components.

Method	Many $\uparrow$	Med. $\uparrow$	Few $\uparrow$	All $\uparrow$
CE	65.12	36.60	9.06	38.32
+FG-CutMix+BFR	65.48	42.09	23.93	44.83 $\uparrow$ 6.51
BCL	66.85	52.92	33.07	51.84
+FG-CutMix	67.11	52.11	38.84	53.38 $\uparrow$ 1.54
+BFR	66.96	53.65	35.86	52.97 $\uparrow$ 1.13
+FG-CutMix+BFR	66.79	53.46	40.34	54.19 $\uparrow$ 2.35

Table 4: Further analysis of CutMix region selection. All variants use the same CutMix pipeline but differ in how the preserved region is selected.

CutMix Variant	Acc. (%)
Standard CutMix (random selection)	52.17
CutMix w/ SAM-based foreground selection	52.43
CutMix w/ CAM-based foreground selection	52.63
FG-CutMix w/ RL-based foreground selection	54.19

standard training (CE), re-balancing techniques (LDAM-DRW [4], BBN [107]), contrastive learning (SBCL [22], BCL [110]), and advanced representation optimization (GBG [43], MKP [8]). By consistently using standard ResNet backbones, this integration fairly and directly validates OFBD’s ability to boost performance by mitigating background bias across diverse learning paradigms.

Implementation. For fairness, we reproduce baselines with official codes and fixed seeds. We use ResNet-32 for CIFAR10/100-LT [4], ResNet-50 [20] for ImageNet-LT and iNaturalist 2018, and ViT-B/16 [14] for cross-architecture evaluation. All models are trained on an RTX 4090 GPU with a batch size of 256. For OFBD, we set $\lambda = 0.5$, $\gamma = 0.8$, $\alpha = 1.0$, and $\beta = 0.02$ across datasets. Extensive ablation studies, further analyses, and implementation details are deferred to Appendix D.

5.2 Effectiveness as a Plug-in Module

CIFAR100-LT and CIFAR10-LT [4]. Table 1 reports results on CIFAR-10/100-LT after integrating OFBD into existing methods. Across 42 experimental settings, OFBD improves 40 of them, with the largest gains of +6.51% on CIFAR-100-LT and +3.88% on CIFAR-10-LT. For the standard CE baseline, OFBD improves CIFAR-100-LT by +6.51%, +3.85%, and +2.51% under $r = 100$, 50, and 10, respectively, and improves CIFAR-10-LT by +3.88%, +3.86%, and +0.86%. For re-balancing methods, OFBD improves LDAM-DRW by up to +3.72% and BBN by up to +2.36% on CIFAR-100-LT. For contrastive-learning methods, OFBD improves BCL and SBCL by up to +2.35% and +1.99%, respectively. For representation-optimization methods, OFBD improves GBG by up to +2.31% and MKP by +1.21% under high imbalance. These consistent gains verify the plug-in effectiveness of OFBD across diverse long-tailed learning paradigms.

The Many/Med./Few statistics on CIFAR-100-LT with $r = 100$ further show that OFBD mainly benefits data-scarce classes while preserving head-class performance. OFBD improves Few-class accuracy by +14.79% on CE, +6.39% on LDAM-DRW, +5.04% on BBN, +7.27% on BCL, +5.22% on SBCL, +7.36% on GBG, and +2.38% on MKP. In contrast, Many-class changes remain small for strong baselines (e.g., -0.06% on BCL, -0.13% on GBG, and +0.29% on MKP). These results indicate that OFBD alleviates tail-class degradation without sacrificing head-class recognition.

ImageNet-LT and iNaturalist 2018 [52, 79]. Table 2 reports results on large-scale long-tailed datasets with both ResNet-50 and ViT-B backbones. With ResNet-50, OFBD consistently improves all baselines on both datasets. The gains are particularly large for the vanilla CE baseline, improving on ImageNet-LT from 44.80% to 47.39% (+2.59%) and on iNaturalist 2018 from 65.95% to 69.76% (+3.81%), showing that OFBD effectively complements standard training. For stronger baselines, OFBD still brings stable gains: +1.14%, +0.88%, +0.51%, and +0.59% on ImageNet-LT for BCL, GCL, GBG, and MKP, and +0.53%, +1.16%, +0.44%, and +0.22% on iNaturalist 2018. With ViT-B, OFBD also improves ViT and DeiT-LT by +1.66%/+0.57% on ImageNet-LT and

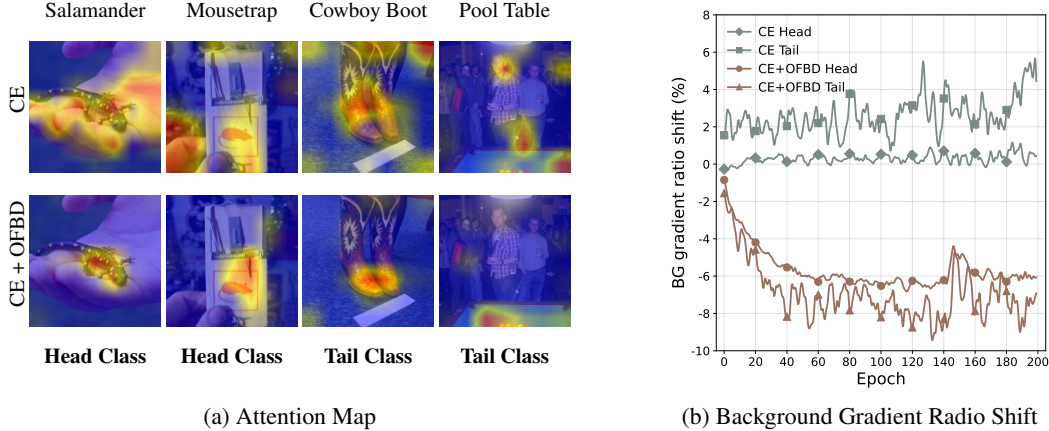


Figure 3: Visualization of background debiasing on CIFAR100-LT ($r = 100$). (a) Pretrained-CAM attention maps of CE and CE+OFBD on three representative head, medium, and tail classes, showing that OFBD reduces background co-activation. More visualization results are provided in Appendix C.1. (b) Background gradient ratio (Eq. 2) shift during training. OFBD reduces optimization-level background bias (Sec. 3.3), especially for tail classes.

+2.39%/+1.64% on iNaturalist 2018, respectively. These consistent improvements across datasets, baselines, and CNN/Transformer backbones verify the generality of OFBD as a plug-in framework.

5.3 Ablation Study and Analysis

In this section, we present key ablation and analysis experiments to characterize the effectiveness, design choices, and debiasing behavior of OFBD. More experiments are provided in Appendix D.

Component Analysis. We conduct component ablation on CIFAR100-LT ($r = 100$) with ResNet-32. We choose CE and BCL [110] as representative baselines: CE reflects standard long-tailed training, while BCL represents a strong contrastive-learning baseline. As shown in Table 3, adding the full FG-CutMix+BFR framework to CE improves the overall accuracy from 38.32% to 44.83% (+6.51%), mainly due to the large Few-class gain from 9.06% to 23.93%. On BCL, FG-CutMix and BFR improve the overall accuracy by +1.54% and +1.13%, respectively. Combining them further improves BCL from 51.84% to 54.19% (+2.35%), with Few-class accuracy increasing from 33.07% to 40.34%. These results show that both components are effective, especially for tail classes.

Analysis of RL-based foreground region selection. Built upon the BCL [110], we further analyze the region selection strategy in FG-CutMix on CIFAR100-LT ($r = 100$). As shown in Table 4, standard CutMix with random regions achieves only 52.17%, indicating that random replacement may preserve irrelevant backgrounds or discard target-related evidence. Using pretrained SAM- and CAM-based foreground regions selection improves the accuracy to 52.43% and 52.63%, respectively, but the gains remain limited. In contrast, our RL-based foreground regions selection achieves the best accuracy of 54.19%, outperforming random, SAM-based, and CAM-based selection by +2.02%, +1.76%, and +1.56%, respectively. These results show that accurate foreground localization is not necessarily the most effective mixing strategy, while RL-based selection better balances target preservation and background replacement. Further ablation analysis is provided in Appendix D.7.

Visualization of Background Debiasing. We further visualize whether OFBD reduces background-biased representation and optimization. As shown in Fig. 3a, CE predictions co-activate target foregrounds and surrounding backgrounds. With OFBD, attention becomes more object-focused, indicating reduced reliance on irrelevant background cues. Fig. 3b shows that CE produces a positive and increasing background-gradient-ratio shift, particularly for tail classes, while CE+OFBD suppresses this shift for both head and tail classes, especially for tail classes. Overall, these results confirm that OFBD successfully mitigates background bias, encouraging the model to focus more on target semantics rather than spurious background cues.

Table 5: Validation of B_u on ImageNet-LT.

Group	B_u -FG↓	B_u -BG↑	BG AUROC↑
Overall	0.3657	0.6321	0.7182
Many	0.3639	0.6340	0.7247
Medium	0.3651	0.6315	0.7161
Few	0.3726	0.6292	0.7049

Table 7: Foreground selection comparison.

Method	Overall↑	Many↑	Med.↑	Few↑
SaliencyMix [78]	52.85	67.62	53.47	34.90
SnapMix [25]	52.01	67.45	53.94	31.75
PuzzleMix [33]	52.52	67.06	53.80	34.06
ResizeMix [62]	52.03	67.24	53.91	32.96
Attentive CutMix [81]	53.19	68.92	53.99	33.91
RL selector	54.21	66.51	54.00	40.10

Table 6: Background intervention analysis on ImageNet-LT.

Method	FG-only↑	BG-only↓	BG-swap↑
BCL [110]	40.18	20.15	31.21
+OFBD	41.99↑ 1.81	18.91↓ 1.24	38.23↑ 7.02

Table 8: Results with different imbalance ratios.

r	Method	Many↑	Few↑	Overall↑
200	BCL [110]	66.60	26.33	46.30
	+OFBD	65.37↓ 1.23	30.08↑ 3.75	47.20↑ 0.90
300	BCL [110]	68.00	26.09	44.62
	+OFBD	65.10↓ 2.90	29.59↑ 3.50	45.78↑ 1.16
400	BCL [110]	68.33	24.04	42.99
	+OFBD	64.60↓ 3.73	27.70↑ 3.66	44.16↑ 1.17

5.4 Further Analysis of Background Debiasing

Background Score Validation. We evaluate whether B_u captures background-related information using CAM-guided SAM masks. Specifically, CAM responses from a pretrained classifier are used as point prompts for SAM to obtain foreground/background masks. As shown in Table 5, B_u assigns higher average scores to background regions than foreground regions and achieves BG AUROC above 0.70 across all groups, demonstrating its ability to identify background-biased features.

Background Intervention. To verify whether OFBD reduces background shortcut reliance, we conduct intervention-based evaluations on ImageNet-LT [52] using foreground-only, background-only, and background-swapped inputs with fixed post-hoc foreground masks. As shown in Table 6, OFBD improves FG-only accuracy by +1.81 points while reducing BG-only accuracy by 1.24 points, indicating a more object-focused representation. Moreover, it achieves a +7.02-point gain under BG-swap, demonstrating improved robustness against background bias under distribution shifts.

Foreground Selection Analysis. We compare the proposed RL-based foreground selector with representative mixing strategies, including SaliencyMix [78], SnapMix [25], PuzzleMix [33], ResizeMix [62], and Attentive CutMix [81]. As shown in Table 7, the RL selector achieves the best overall accuracy and Few-shot performance, improving Few-shot accuracy by +5.20 points over the strongest heuristic baseline, demonstrating the effectiveness of adaptive foreground preservation for long-tailed recognition. This improvement highlights the advantage of learning region selection over manually designed heuristic strategies.

Performance under Different Imbalance Ratios. We further evaluate OFBD under more severe imbalance settings by increasing the imbalance ratio on CIFAR-100-LT. As shown in Table 8, OFBD consistently improves overall and Few-shot accuracy across different imbalance ratios. Specifically, OFBD improves overall accuracy by +0.90/+1.16/+1.17 points under $r = 200/300/400$, while achieving +3.75/+3.50/+3.66 points gains on Few-shot classes. These results demonstrate that OFBD remains effective under increasingly challenging long-tailed distributions.

6 Conclusion

Revisiting long-tailed recognition via background bias, we reveal that tail degradation arises from sample scarcity combined with background-biased representation and optimization (i.e., larger distribution shifts and gradient reliance). To mitigate this, we propose Object-Focused Background Debiasing (OFBD), a plug-in framework. Specifically, FG-CutMix preserves target foregrounds while diversifying backgrounds, and parameter-free BFR suppresses background-biased features. Experiments across four datasets demonstrate that OFBD improves mainstream methods, boosting tail accuracy without sacrificing head performance on both CNN and Transformer backbones. Its plug-in design inspires extensions to other long-tailed tasks like object detection and trajectory prediction.

Limitation: the performance of OFBD on domains with highly complex scenes (e.g., medical imaging datasets) remains to be explored.

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